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Revealing mechanism of micro-aeration for enhancing volatile fatty acids production from swine manure

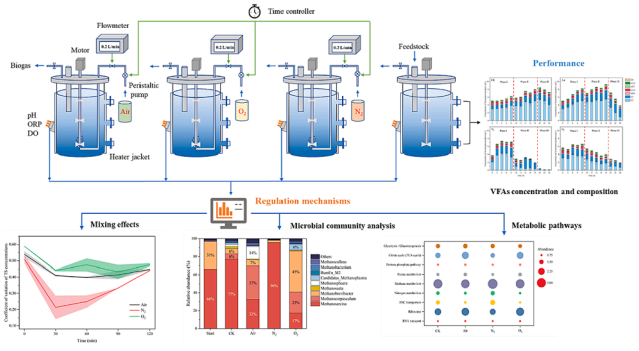
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HIGHLIGHT

- Air-micro-aeration had significant improvement with VFAs increased by 22.4%.
- Air-micro-aeration at ORP +5.5 mV higher than reference ORP was favorable to VFAs.
- Different gas micro-aeration did not achieve uniform mixing.
- Air-micro-aeration promoted electron transfer activity and energy generation.
- Consumption and conversion of VFAs by methanogens were slow by Air-micro-aeration.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Micro-aeration is considered a new strategy for improving volatile fatty acids (VFAs) production of agricultural waste. This study investigated the effect and mechanism of micro-aeration of air and oxygen (O_2) on VFAs production from swine manure. The results showed that Air-micro-aeration had the most significant improvement effect, with the highest VFAs of 8.21 g/L, which was increased by 22.4%. Moreover, the mixing effects of different micro-aeration were limited, and the microbial communities significantly varied. *Firmicutes* and *Bacteroidota* were the dominant hydrolytic and acidogenic bacteria, and Air-micro-aeration preferentially promoted electron transfer activity and energy generation. *Methanosarcina*, *Methanocorpusculum*, and *Methanobrevibacter* can adapt to environmental changes according to their different oxygen tolerance, and the consumption and conversion of VFAs by methanogens were slow under Air-micro-aeration condition. This study revealed mechanism of micro-aeration for improving VFAs production from swine manure, providing a theoretical basis for micro-aeration regulation optimization.

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1. Introduction

Micro-aerobic environment is a state between aerobic and anaerobic conditions with a low dissolved oxygen (DO) concentration, that is created by injecting a trace amount of air or oxygen (O_2) (DO range of 0.1-1.0 mg/L) into an anaerobic bioreactor (Wu et al., 2015; Chen et al., 2020; Zhang et al., 2021). Recent studies have shown that micro-aeration is an effective approach that facilitates anaerobic digestion (AD), such as improving hydrolysis rate and enhancing methane production (Nguyen and Khanal, 2018; Wang et al., 2020). The hydrolysis stage is generally regarded as a rate-limiting step of AD, especially when recalcitrant lignocellulosic biomass, such as straw, are used, and hydrolysis rates under aerobic and anoxic conditions are significantly higher than those under anaerobic conditions (Shrestha et al., 2017; Wang et al., 2020).

Given that AD system has a certain tolerance to O_2, micro-aeration can have a positive impact on microbial metabolism and energy transfer. Lim et al. (2014) studied the co-digestion characteristics of brown water and food waste at a micro-aeration intensity of 5 and 7 mL O_2 $L^{-1}d^{-1}$, and the results showed that micro-aeration enhanced microbial diversity and volatile fatty acids (VFAs) production, and the hydrolysis rate and methane yield also increased. Microbial analysis indicated that micro-aeration can promote the proliferation of diverse bacterial populations and increase the proportion of *Firmicutes* clones related to hydrolysis in contrast to anaerobic conditions (Lim et al., 2014). Yu et al. (2020) found that hydrolytic/fermentative bacteria and acetoclastic methanogens (e.g., *Methanosaeta*) were in higher abundances with controlled micro-aeration (5 and 10 mg O_2/L-reactor), which facilitated syntrophic interactions between microorganisms. The relative abundance of dominant bacterium *Firmicutes* increased from 58% to 72%, and the relative abundance of methanogenic archaea *Methanosarcina* and *Methanobacterium* with certain oxygen tolerance also increased (Fu et al., 2016).

VFAs are the intermediate products and can be consumed at the methanogenic stage in AD. When air or O_2 is introduced into an anaerobic reactor, the proliferation of strictly anaerobic methanogens is inhibited to a certain extent due to the enhanced activity of facultative hydrolysis bacteria and fermentation bacteria, and thus VFAs may accumulate. Lim and Wang et al. (2013) reported that VFAs concentration increased from 1.69 g COD/L to 5.57 g COD/L after 4 days of micro-aeration pretreatment, and methane production increased by 21%, but long-term experiment is needed to verify. It was reported in a previous study that the rate of methane production increased by 19.6%, through the mass-transfer (relative contributions of 62.9%) and micro-aerobic environment (relative contributions of 37.1%) (Yang and Deng, 2020). To prevent excessive VFAs accumulation under high OLR conditions, Nguyen et al. (2019) found that VFAs concentrations decreased from 9.84 g COD/L to 3.21 g COD/L at -470 mV after conducting intermittent ORP-based micro-aeration control. To date, most studies on micro-aeration in AD aimed at improving methane production. In the high-rate production of short chain carboxylic acids (SCCAs, including VFAs and lactic acid), Cao et al. (2022) confirmed that micro-aeration enhanced VFAs by 20.3% and shortened the time to reach the maximum from 18 days to 10 days, but the effect of micro-aeration on VFAs production requires more studies. The mechanism of micro-aeration promoting the production of VFAs still remains unclear. The reasons could be that micro-aerobic environment enhances metabolic activities of hydrolytic bacteria and acid-producing bacteria, or the gas mixing improves the mass transfer of organic matters.

In this study, the effect and mechanism of micro-aeration on VFAs production were explored by comparing different micro-aeration (Air and O_2) with N_2 addition. It is expected to provide a theoretical basis for the optimization of VFAs production parameters and process improvement of subsequent micro-aeration regulation, and a novel strategy for realizing an environment-friendly, economic and efficient acid production process.

Materials and methods

1.1. Feedstock and inoculum

Swine manure was collected from a pig farm in Changping, Beijing, and stored in a freezer at -20 °C before the use. The anaerobically digested sludge as inoculum was collected from an anaerobic reactor stably operated for a long time in the laboratory with a previously described method and heat-treated at 85 °C for 3 h (Cao et al., 2021). The physicochemical properties of swine manure and inoculum are shown in Table 1. N_2 (purity $\geq$ 99.9%) and O_2 (purity $\geq$ 99.2%) gases were purchased from Jinggao Gas Co., Ltd (Beijing). Room air (N_2:79.7%, O_2:20.2%, CO_2:0.1%) was directly used for the micro-aeration of air source.

1.2. Experimental design

Experiments were performed using four sets of 6.5 L stainless steel batch reactors with effective volumes of 6.0 L. The reactor was equipped with pH, ORP, DO, and temperature sensors. For sample collection and testing of fermentation broth characteristics, the upper, middle, and lower sampling ports were set at different heights (10, 20 and 30 cm) above the bottom of the reactors. A 10 L gas collection bag was attached to the reactor for gas collection. The schematic diagram of micro-aeration-based AD system is shown in Supplementary Material. In this study, swine manure as the sole substrate was added to each reactor with a VS concentration of 4% and a substrate/inoculum ratio (v/v) of 9:1. The initial pH was adjusted to 10.0±0.2 by using 6.0 M NaOH, and the temperature was maintained at 35±1 °C by the outer wall of the reactor with a circulating water bath according to previous studies (Zhang et al., 2020). A micro-aeration intensity of 1000 mL/L/d was conducted as previously described (Cao et al., 2022) and the micro-aeration sources were air and O_2. ORP and pH changes during the experiment are shown in Supplementary Material. The experimental period was 24 days, and about 5 mL of liquid samples were collected from each reactor every 2 days for analysis of characteristics.

1.3. Analytical methods

Total solids (TS) and volatile solids (VS) of swine manure and inoculum were measured according to the standard methods (APHA, 2005). C and N contents were determined with an elemental analyzer (Perkin Elmer 2400 SeriesII, Sheffied, UK). pH and ORP were measured with a pH meter (Easyferm Plus PHI S8 120, Hamilton, Switzerland) and an ORP meter (Oxytrode Platinum, Hamilton, Switzerland), respectively. Crude fiber was analyzed using a cellulose analyzer (ANKOM 220i, USA) following the Van Soest method (Soest, 1963). The volume and composition of the biogas were determined with a mass flow controller (ACU10FD-LC, ACCU, Beijing) and a portable biogas analyzer (Biogas 5000, Geotech, UK). VFAs were analyzed as described by Cao et al. (2021).

Table 1
Physicochemical characteristics of swine manure and inoculum

Items	Swine manure	Inoculum
TS (% FM)	30.7±0.4	10.6±0.1
VS (% FM)	24.4±0.4	6.6±0.0
pH	8.62±0.03	7.30±0.03
VFAs (g/L)	ND	1.77±0.18
Crude protein (%, TS)	22.05±0.23	ND
Crude fiber (%, TS)	37.04±0.18	ND
C (% TS)	39.3±0.6	25.6 ± 0.2
N (% TS)	3.4±0.1	3.2 ± 0.2
C/N	11.4±0.3	8.1 ± 0.5

Note: Data include the means ± standard deviation (n = 3), ND: not detected, FM: fresh matter, TS: total solids, VS: volatile solids.

To analyze the structure of microbial communities, approximately 5 mL of samples were obtained from each reactor every 6 days. High-throughput sequencing was performed by Allwegene Technology Co. Ltd (Beijing, China) using an Illumina MiSeq platform (Illumina Lnc., CA, USA). The primer pairs 338F-806R and 344F-806R were used to amplify bacterial 16S rRNA and archaeal 16S rRNA genes, allowing the original sequence data to be generated. The functional characteristics of the microbial communities were analyzed by phylogenetic investigation of communities by reconstruction of unobserved states (PICRUSt), and the metabolic pathways were classified and generated based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) database.

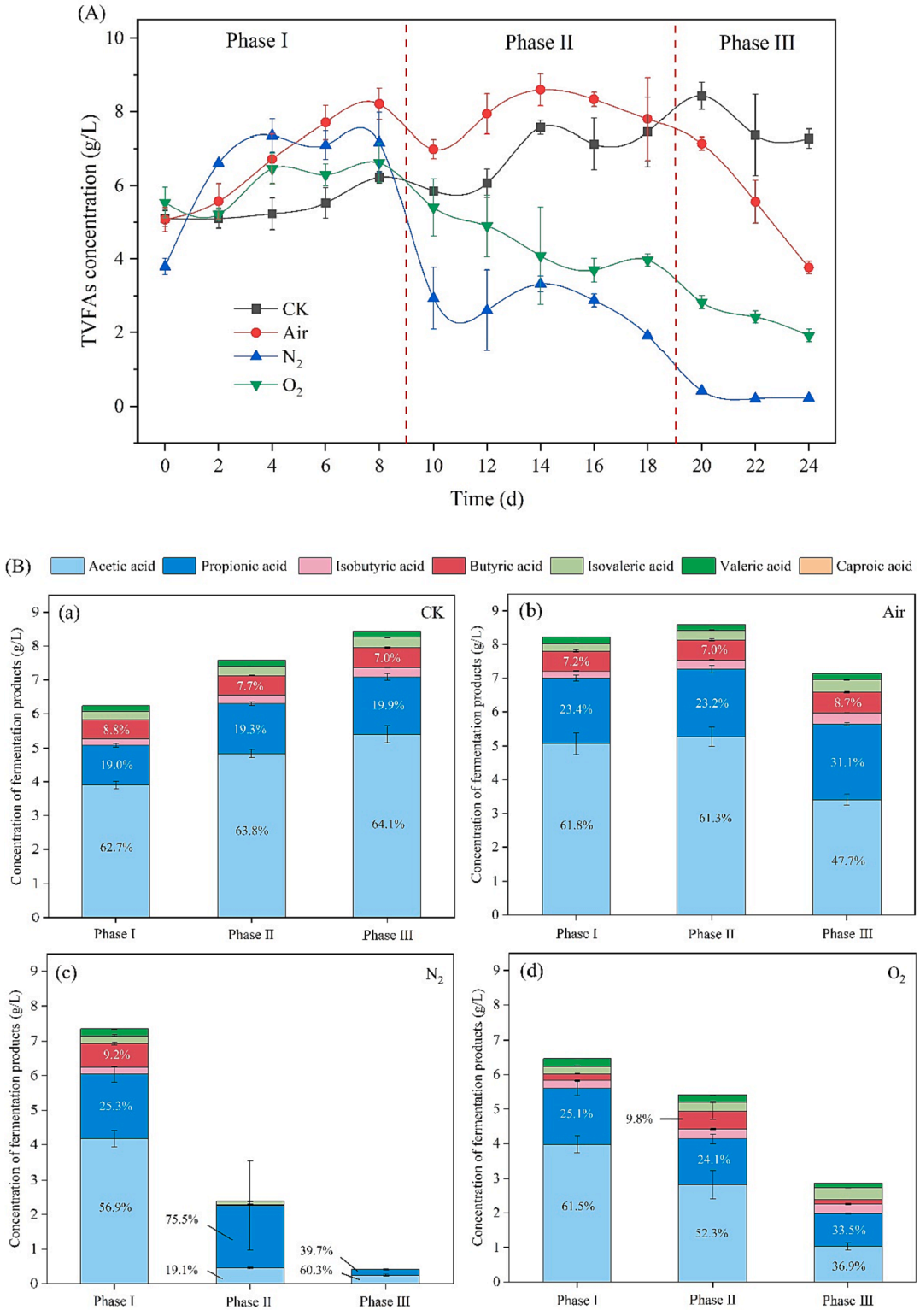


Fig. 1. Changes in TVFAs concentration (A) and composition distribution (B) at the maximum concentration in 3 Phases under the conditions of (a) CK, (b) Air-micro-aeration, (c) N_2 addition, (d) O_2-micro-aeration.

1.4. Data processing and statistical analysis

All data were calculated using Microsoft Office Excel 2019. Data statistical analysis was performed using IBM SPSS Statistics 25, and the significance level was 0.05. All figures in this study were plotted using Origin 2021.

2. Result and discussion

2.1. Total Volatile fatty acids concentration changes under different micro-aeration conditions

Changes in TVFAs concentration under different micro-aeration conditions were shown in Fig. 1A. According to the variation in TVFAs concentration under the Air-micro-aeration condition, the 24-day AD process was divided into three phases: Phase I: In the first 8 days, the concentration of TVFAs gradually accumulated; Phase II: From day 8 to day 18, the concentration of TVFAs remained stable; Phase III: After day 18, the AD system entered the methanogenic stage, and the concentration of TVFAs decreased rapidly.

In Phase I, the concentration of TVFAs demonstrated an upward trend under different micro-aeration conditions. The highest concentration of TVFAs was achieved through Air-micro-aeration, and the maximum value was 8.21 g/L. Under N_2 addition condition, the TVFAs concentration reached a maximum value (7.36 g/L) on day 4, and then remained stable at about 7.00 g/L. The highest TVFAs concentration by O_2-micro-aeration was 6.62 g/L. For the improvement of TVFAs production, O_2-micro-aeration was lower than that of Air-micro-aeration and N_2 addition, but higher than that of without micro-aeration. Compared with AD, Air-micro-aeration, N_2 addition, and O_2-micro-aeration improved TVFAs production by 22.4%, 17.8%, and 10.8%, respectively. In Phase II, Air-micro-aeration reactor maintained a stable acidification stage, with the highest and average TVFAs of 8.59 and 7.93 g/L, respectively. However, the biogas of N_2 addition and O_2-micro-aeration reactors gradually increased (See Supplementary Material), and the average TVFAs dropped to 2.73 and 4.41 g/L, respectively. N_2 addition enhances the mass transfer of substrate and fosters the growth and activities of methanogens (Wang et al., 2020). Therefore, more VFAs (mainly acetic acid) were converted to CH_4 (See Supplementary Material) to make the VFA production relatively lower. Under anaerobic condition, TVFAs accumulated continuously, and the maximum concentration was 7.58 g/L. In Phase II, given that N_2 addition and O_2-micro-aeration reactors were still at the methanogenic stage, TVFAs continued to decrease. Air-micro-aeration reactor began to enter the methanogenic stage, and TVFAs decreased rapidly. The TVFAs of the anaerobic reactor had a maximum of 8.43 g/L on day 20, but the concentration began to decrease due to the AD system entered the methanogenic stage.

ORP is commonly used to measure the oxidative (more positive values) or reductive (more negative values) environment in an aqueous phase, and Nguyen et al. (2019) developed an ORP-controlled intermittent micro-aeration system to improve the methane yield (ORP at +25 mV from the baseline ORP). In this study, compared with anaerobic conditions, N_2 addition, Air-, and O_2-micro-aeration increased ORP by about 1.5, 5.5, and 11.0 mV, respectively. O_2-micro-aeration and Air-micro-aeration had a more significant improvement in ORP than N_2 addition. The relationship between relative ORP and TVFAs were shown in Fig. 2A. It was observed that Air-micro-aeration at an ORP +5.5 mV higher than the anaerobic ORP was favorable to TVFAs, especially acetic acid production, while O_2-micro-aeration at an ORP +11.0 mV higher than the anaerobic ORP caused a significant reduction of TVFAs due to excess aerobic respiration. The same phenomenon was also observed in the AD of blackwater by micro-aeration, and the results indicated that the TVFAs increased from 90-269 mg/L to 339-396 mg/L with the oxygen dosage increased from 0 (ORP=-308 mV) to 50 mg/g COD_{feed} (ORP=-109 mV), while the TVFAs decreased to 203-304 mg/L with the oxygen dosage continued to expand to 150 mg/g COD_{feed} (ORP=-69 mV) (Guo et al.,2021). Intermittent micro-aeration at an ORP +25 mV higher than the baseline ORP accelerated the VFAs conversion rapidly, while when ORP +50 mV caused significant accumulation of VFA due to inhibition of methanogens as indicated by reduced methane yields (Nguyen et al., 2019).

2.2. Total Volatile fatty acids composition distribution under different micro-aeration conditions

As shown in Fig. 1B, acetic acid was the main component of VFAs, followed by propionic acid under different micro-aeration conditions in Phase I. Under the conditions of no micro-aeration (CK), Air-micro-aeration, N_2 addition and O_2-micro-aeration, the proportions of acetic acid at the maximum VFAs concentration were 62.7%, 61.8%, 56.9%, and 61.5%, respectively, and the proportions of propionic acid were 19.0%, 23.4%, 25.3%, and 25.1%, respectively (Fig. 1B). The result showed that micro-aeration had no significant effect on TVFAs composition distribution in the early stage ($P > 0.05$).

From day 8 to day 18, the TVFAs composition distribution under no micro-aeration, Air-micro-aeration and O_2-micro-aeration conditions was similar to that in the first 8 days. However, under N_2 addition

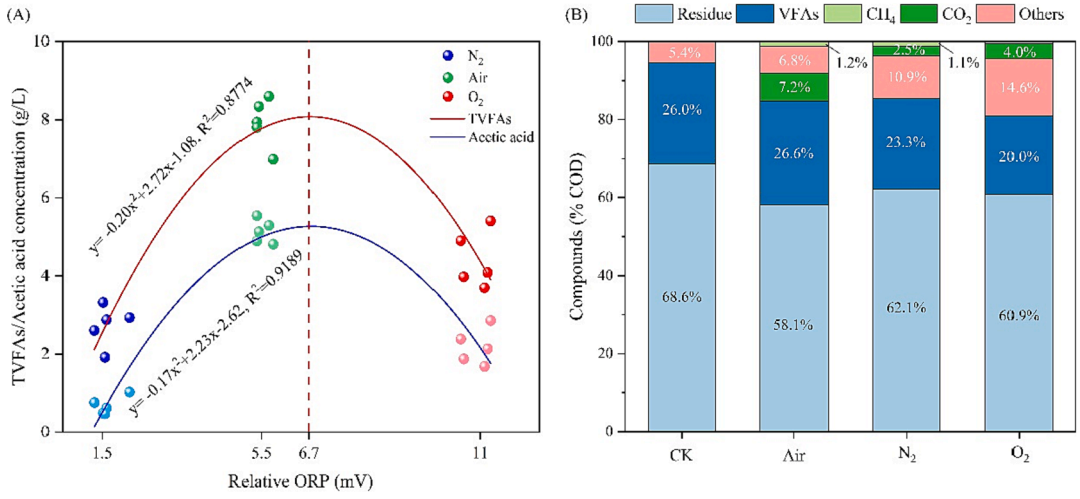


Fig. 2. Correlation analysis between TVFAs, acetic acid and relative ORP (A), Day 10-Day 18; Carbon mass balance at the peak of VFAs production (B), CK: Day 20; Air: Day14; N_2: Day 4; O_2: Day 4.

condition, the proportion of acetic acid was only 19.1%, and the proportion of propionic acid increased to 75.5% at the peak of TVFA production (Fig. 1B). The reason was that the AD system rapidly reached the methanogenic stage under N_2 addition condition.

After day 18, the percentages of acetic acid and propionic acid were 64.1% and 19.9% (Fig. 1B), respectively, and no significant change in TVFAs components was found in the whole AD process without micro-aeration ($P>0.05$). Due to Air-micro-aeration reactor began to enter the methanogenic stage, acetic acid was rapidly converted into biogas, and the concentration decreased from 5.65 g/L (61.3%) to 3.20 g/L (47.7%). Compared with acetic acid, propionic acid was not easily converted into methane, and remained stable in Phases II and III, and its concentrations were 2.14 and 2.20 g/L, respectively. However, owing to a decrease in acetic acid content, the proportion of propionic acid increased from 23.2% to 31.1%. Methane production in the N_2 addition reactor was nearly completed and the concentrations of acetic and propionic acid were only 0.22 (60.3%) and 0.15 g/L (39.7%), respectively.

The results indicated that micro-aeration did not change the TVFAs composition distribution, and acetic acid was the main component. The TVFAs composition distribution is mainly related to the bacterial community. There was no significant change in the bacterial community (see Supplementary Material), and *Firmicutes* and *Bacteroidota* were the dominant hydrolysis and acidogenic bacteria under different micro-aeration conditions.

2.3. Carbon mass balance at the peak of volatile fatty acids production

The results of carbon mass balance at the peak of VFAs production under different micro-aeration conditions are shown in Fig. 2B. The carbon content in the residue was 68.6% without micro-aeration, and the hydrolysis rate was lower than the hydrolysis rates under micro-aeration conditions. Wang et al. (2020) investigated the effects of different nanobubble water (NBW) on the AD of cellulose for methane production, and the results showed that the rate of total organic carbon removal reached 31.8%, 34.1%, 34.6%, and 34.9% in deionized water, Air-NBW, N_2-NBW, and O_2-NBW, respectively, indicating that NBW enhanced cellulose hydrolysis. In the present study, given that no biogas production was observed in the anaerobic reactor (CK) before the VFAs reached the maximum, most of the soluble organic carbon (82.8%) was retained in the form of VFAs. Different from other reactors, the Air-micro-aeration reactor involved exogenous carbon due to the introduced CO_2, which may be a factor that should be considered in calculating carbon mass balance. Until the VFAs reached the maximum levels in the Air-micro-aeration reactor (Day 14), the carbon content of introduced CO_2 was only 0.07 g, whereas the carbon content of the substrate was about 94.32 g. In terms of accumulated biogas, the carbon content in the form of CH_4 and CO_2 was 1.17 and 6.82 g, respectively, and thus exogenous carbon that entered the reactor through Air-micro-aeration was not considered. The highest hydrolysis rate was obtained under the Air-micro-aeration condition and increased by 15.3% relative to that in No-micro-aeration. The carbon content of the hydrolysate was 63.6% for VFAs production, 20.2% for biogas, including CH_4 and CO_2, and 16.2% for other forms. N_2 addition and O_2-micro-aeration did not improve the hydrolysis rate of fermentation materials, but the maximum VFAs level was lower than that under Air-micro-aeration or anaerobic condition. The results showed that the hydrolyzed organic carbon was not effectively transformed into VFAs and converted into biogas and other soluble substances (proteins, polysaccharides, amino acids, structural components of cells, etc.), especially microbial biomass. The carbon content in other forms under N_2 addition and O_2-micro-aeration conditions was 10.9% and 14.6%, respectively. In this study, the same amount of O_2 gas and air were injected into the reactors daily and the reactor with O_2 gas received approximately four times more oxygen than that with air gas, since air contained about 21% oxygen gas. So, O_2-micro-aeration has a higher aeration intensity than Air-micro-aeration. A high aeration intensity is favorable for microbial growth, and the total soluble hydrolysate used for biomass generation is as high as 18.5%, which is only 0.9% and 6.6% under anaerobic and appropriate micro-aeration conditions, respectively (Xu et al., 2014). Electron transport system activities (ETS) generally occurs in the bacterial cell membrane (Szabo, 2003). The highest ETS activity of O_2-NBW, N_2-NBW and Air-NBW reactors increased by 1.7, 1.6 and 1.1 times compared to the control (Wang et al., 2020). ETS can transfer electrons from the coenzymes (NADH, NADPH) produced by the tricarboxylic acid (TCA) cycle to the terminal electron acceptor in the oxidative phosphorylation (electron transport) pathway. The surface of NBs has negative charges and can attract NAD^+ (Kyzas et al., 2019); NBs also have a higher movement speed, favoring the movement of NAD^+, which is used for the reduction reaction to generate NADH. Therefore, although N_2 addition and O_2-micro-aeration increased substrate degradation, it was a carbon loss for VFAs production due to the hydrolysate was not conducive to VFAs accumulation.

2.4. Microbial community analysis

2.4.1. Bacterial phyla

The observed_species index represents the number of OTUs actually observed as sequencing depth increases. Under the Air-micro-aeration, N_2 addition and O_2-micro-aeration conditions, the observed_species index increased gradually and then decreased after reaching the peak (Fig. 3). N_2 addition and O_2-micro-aeration increased the number of bacteria rapidly, which reached its peak on day 6. The observed_species index under Air-micro-aeration and anaerobic conditions increased slowly, but the peaks were higher than those under N_2 addition and O_2-micro-aeration conditions. The observed_species index was positively correlated with VFAs production ($P < 0.05$), indicating that the increase in bacterial number was beneficial to VFAs production.

There was no significant change in the bacterial community structures at the class level under different micro-aeration conditions (see supplementary material). The dominant bacterial communities were *Clostridia* and *Bacilli* (Fig. 4A), respectively, and the sum of their relative abundance accounted for about 80% throughout the experiment. *Clostridia* and *Bacilli* belong to *Firmicutes*, which can produce extracellular enzymes (such as cellulases, lipases, and proteases) that play key roles in the metabolism of lignocellulose, proteins and lipids (Lim et al., 2014). *Clostridia* belongs to anaerobic gram-positive bacteria, most of which can convert various carbon sources, including complex lignocellulose, into VFAs (mainly acetate and butyrate), propanol, and butanol (Chai et al., 2019b; Diallo et al., 2021). *Bacilli* widely exist in compost and soil and degrade lignocellulose and protein to produce butyrate (Zalma and El-Sharoud, 2021). *Bacilli* and *Bacteroidia* possess the *buk* pathway, while *Clostridia* can produce butyrate via both the *buk* and *but* pathways (Chai et al., 2019a). In addition, *Bacteroidia* utilize anaerobic respiration for glycolysis to produce acetate and isovalerate (Joyce et al., 2018). It is one of the classes of anaerobic bacteria with the highest resistance to oxygen, maintaining a high relative abundance during the whole AD process of this study.

2.4.2. Archaeal genus

As shown in Fig. 4C, *Methanosarcina* and *Methanobrevibacter* were the dominant archaeal genus at the beginning, accounting for 66% and 31%, respectively. After micro-aeration, the archaeal community structure at the genus level significantly varied. Under anaerobic condition, *Methanosarcina* became the dominant genus, with a relative abundance of 77%. *Methanosarcina* has the ability to use both acetate and CO_2/H_2 to produce methane (De Vrieze et al., 2012). The abundance of *Methanosarcina* decreased to 32% after Air-micro-aeration, and *Methanocorpusculum* increased to become the dominant genus, with a relative abundance of 37%. Under N_2 addition condition, the relative abundance of *Methanosarcina* was as high as 96%. However, the relative abundance of *Methanosarcina* decreased significantly to 17% through O_2-micro-

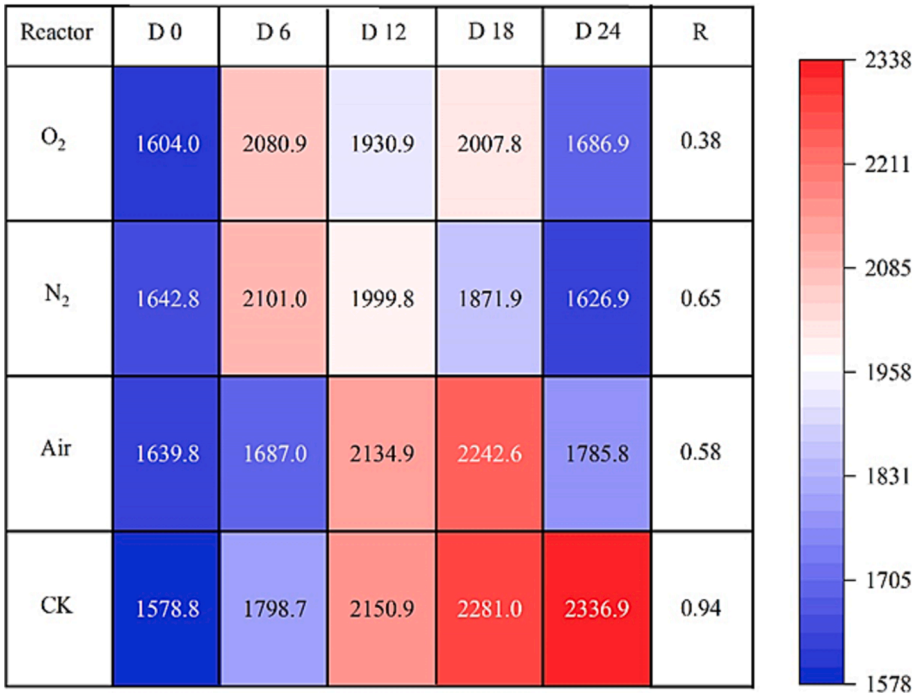


Fig. 3. The relationship between observed_species and VFAs production under different micro-aeration conditions.

aeration, and the relative abundance of *Methanobrevibacter* and *Methanocorpusculum* increased to 45% and 23%, respectively. *Methanobrevibacter* (Leadbetter and Breznak, 1996) and *Methanocorpusculum* (Gilmore et al., 2017) are strictly hydrogenotrophic methanogens (CH_4 production via CO_2/H_2). Nguyen et al. (2019) also observed a similar phenomenon that the shift of dominant archaeal genus from *Methanosarcina* to *Methanobrevibacter* and *Methanocorpusculum* have occurred during ORP-based micro-aeration. *Methanocorpusculum* showed certain oxygen tolerance, was able to adapt to an aerobic environment, and maintained a high relative abundance. *Methanobrevibacter* showed strong oxygen tolerance and became the dominant bacterium with the highest relative abundance under O_2-micro-aeration condition. These results implied that the oxygen tolerance of *Methanosarcina*, *Methanocorpusculum*, and *Methanobrevibacter* increased in turn.

2.5. Preliminary analysis on regulation mechanisms

Three sampling ports were set at different heights (10, 20 and 30 cm) above the base of the reactor for TS analysis. The TS concentration of the fermentation broth in the reactor was measured before and after micro-aeration. The mixing degree of the fermentation broth was assessed according to the coefficient of variation (CV) of TS concentration (Yang and Deng, 2020) and the results are shown in Fig. 5. In order to explore the mixing effect of micro-aeration, the mechanical stirring in the reactors was not started during the whole AD process. Many solids settled in the lower layers of the reactors, resulting in a large CV value of about 0.55. The CV values all decreased in different micro-aeration processes, and a minimum of about 0.20 was obtained by N_2 addition. The minimum CV value of Air-micro-aeration was about 0.45, which was slightly lower than that of O_2-micro-aeration. Different micro-aeration processes with same intensity had different effects on the degree of mixing possibly due to difference in solubility among different gas sources in water. Under normal temperature and pressure, the volume solubility of N_2, Air and O_2 in water are 0.016, 0.019, and 0.031, respectively. N_2 has the lowest solubility in water and cannot be utilized by microorganisms, and thus N_2 addition had the greatest impact on the degree of mixing. Compared with N_2, O_2 has higher solubility in water and can be consumed and utilized by facultative bacteria or removed by reducing substances in reactors. Therefore, O_2-micro-aeration had the least impact on the degree of mixing.

Although the solubility of air in water was similar to that of N_2 and lower than that of O_2 in water, the components in air (mainly O_2 and CO_2) can be consumed and utilized by microorganisms. Thus, the influence on the degree of mixing was less than that of N_2. Gas solubility might not be the major reason for the difference in mixing degree, indicated by the similar mixing degree for air and O_2 (Fig. 5). Other reasons impacting the mixing degree possibly lie in the consumption and utilization rate of gas components (especially O_2) by microorganisms. In addition, due to the different physicochemical properties, different gas types can generate different bubble stabilities and structures at the gas–liquid interface (Zhang et al., 2021). For example, the bubble stability of different gases followed the order H_2>air>N_2>CO_2 (Wang et al., 2019). After micro-aeration was stopped, the solid components in the reactor gradually settled, and the CV value of TS concentration increased. At 120 min, the CV value in different micro-aeration reactors returned to the same level. Among these micro-aeration methods, N_2 addition had the greatest impact on the degree of mixing. However, compared with the definition of uniform mixing in Yang et al. (2021) and Wu (2012), the CV value was less than 0.02 or 0.05. In this study, the minimum CV value (about 0.20) was achieved by N_2 addition, which was much greater than the CV value of uniform mixing. Therefore, the mixing effect of micro-aeration in this study was limited, and the reason for the difference in the mixing effect of different micro-aeration reactors was mainly caused by the nature of the gas source itself. Different from this study, Yang and Deng (2020) reported that air and biogas mixing modes achieved the same mixing efficiency and all improved the mass transfer effect when the aeration intensity was calculated according to the same power input of mechanical mixing. Therefore, when the contribution of improved mass transfer and the reaction of microaerobic environment to methane production were considered, the different mixing effects of the gas source were ignored due to the high aeration intensity. In fact, the mixing effect of micro-aeration is related to the aeration rate or intensity.

Microorganisms in the AD system have considerable oxygen tolerance due to several mechanisms. On the one hand, facultative bacteria with rapid oxygen consumption capacities can consume dissolved

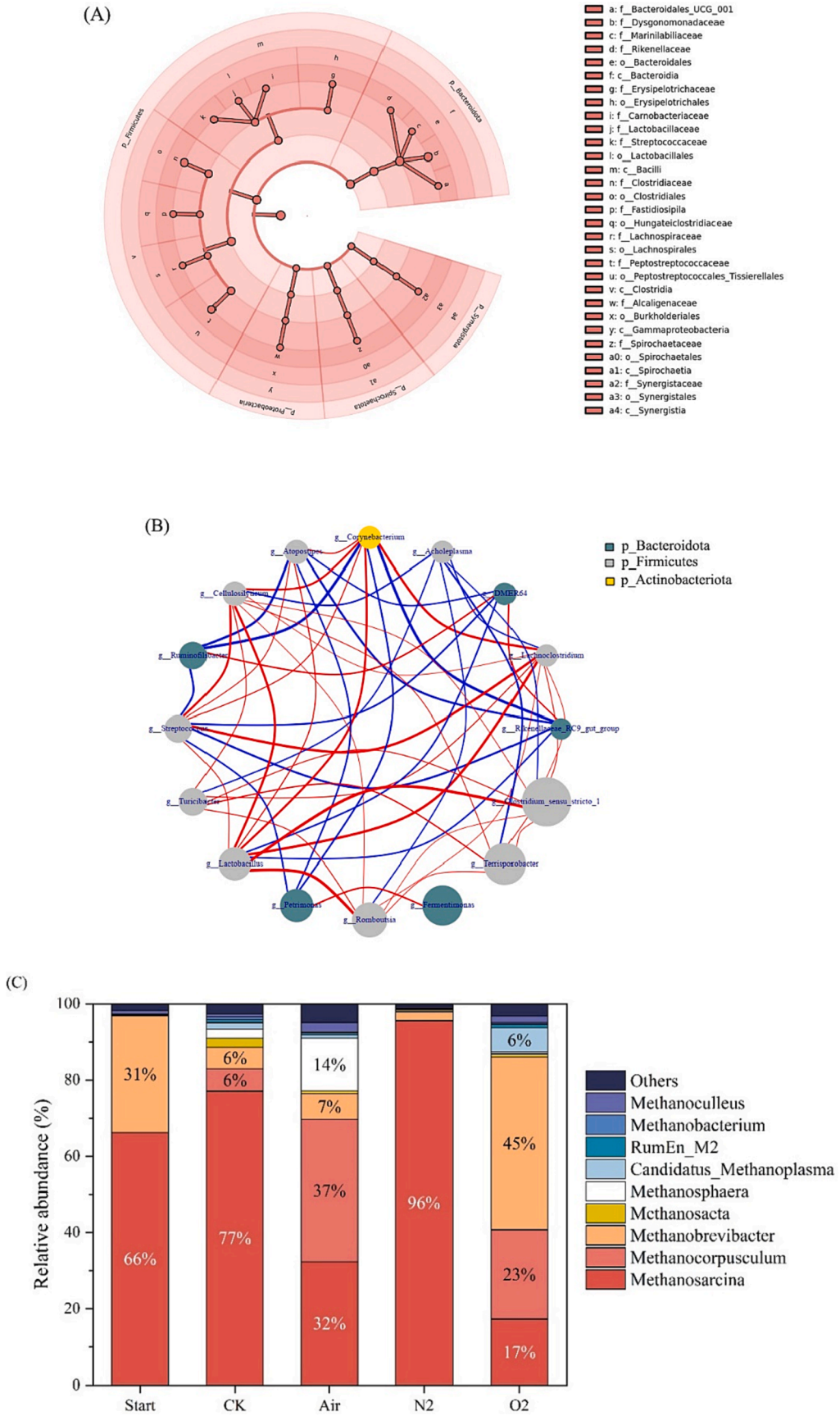


Fig. 4. Bacterial community structures (A) and relationship network of dominant genera (B), and Archaeal community structure at the genus level (C) under different micro-aeration conditions.

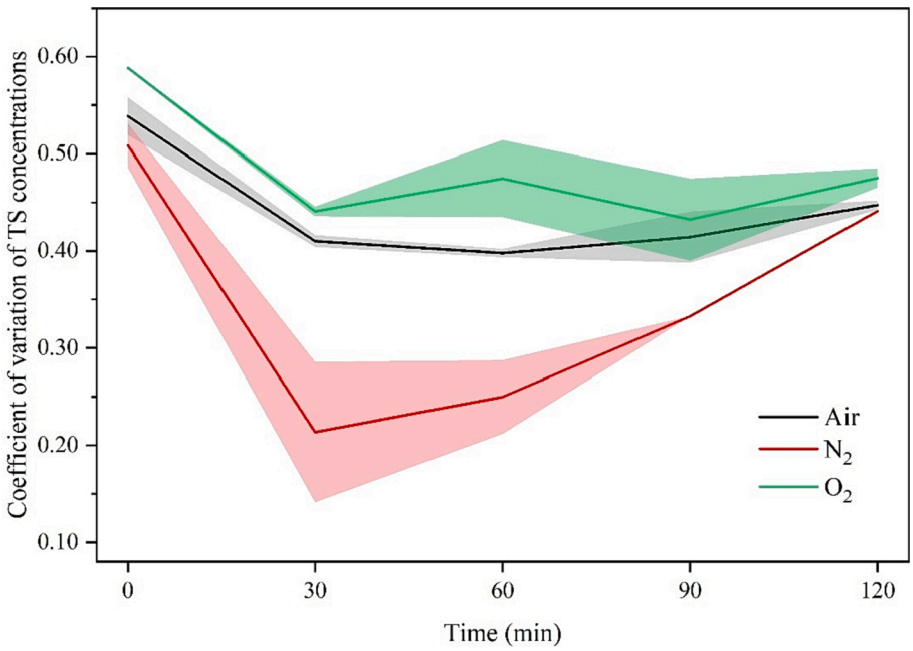


Fig. 5. Changes in the coefficient of variation (CV) of TS concentrations in the different micro-aeration reactors, micro-aeration was 0-30 min, and natural sedimentation after micro-aeration was 30-120 min.

oxygen, which may ensure that obligate anaerobic bacteria in anaerobic environment can survive and remain active under micro-aeration conditions without or slight negative effects (Brioukhanov and Netrusov, 2007; Sheets et al., 2015). This finding was confirmed by slight changes in bacterial community structures in different micro-aerations and anaerobic conditions in this study (see Supplementary Material). On the other hand, some methanogens have a certain level of oxygen tolerance, but the oxygen tolerance of different methanogens varies (Brioukhanov et al., 2006; Horne and Lessner, 2013; Angle et al., 2017). *Methanosarcina* and *Methanobrevibacter* are known to produce anti-oxidative enzymes (such as superoxide dismutase (SOD), catalase, and peroxidase), which enable these archaea to adapt to micro-aeration condition and exhibit high oxygen tolerance (Nguyen and Khanal, 2018). Fu et al. (2016) investigated the impact of micro-aeration on microbial community structure, and the relative abundance of oxytolerant *Methanosarcina* and *Methanobacterium* doubled in the micro-aerobic environment compared with that under anaerobic condition. In the present study, *Methanosarcina*, *Methanocorpusculum*, and *Methanobrevibacter* exhibited different levels of oxygen tolerance, and methanogenic archaea showed survival mechanisms that can adapt to changes in oxygen levels in the environment by adjusting their community structures according to different levels of oxygen tolerance.

The metabolic functions based on the KEGG pathway databases exhibited differences among different micro-aeration conditions, although the magnitude of difference was relatively low (Fig. 6). Genes related to glycolysis/gluconeogenesis, the citrate cycle (i.e., TCA cycle), pentose phosphate pathway, and purine metabolism, which are directly related to the efficiency of hydrolysis and acidification, were more abundant under the Air-micro-aeration condition (Zhang et al., 2022). Air or oxygen bubbles with negative charges promote nicotinamide adenine dinucleotide (NAD^+) transfer, thereby increasing the production of the reduced form nicotinamide adenine dinucleotide (NADH) (Kyzas et al., 2019). Therefore, redox balance can be maintained by regenerating NADH from fermentation intermediates, such as VFAs, which produce small amounts of ATP (Zhang et al., 2021). A trace amount of oxygen was introduced into the anaerobic bioreactor as the terminal electron acceptor to accelerate the hydrolysis of complex organic matter by facultative bacteria and enhance the electron transfer

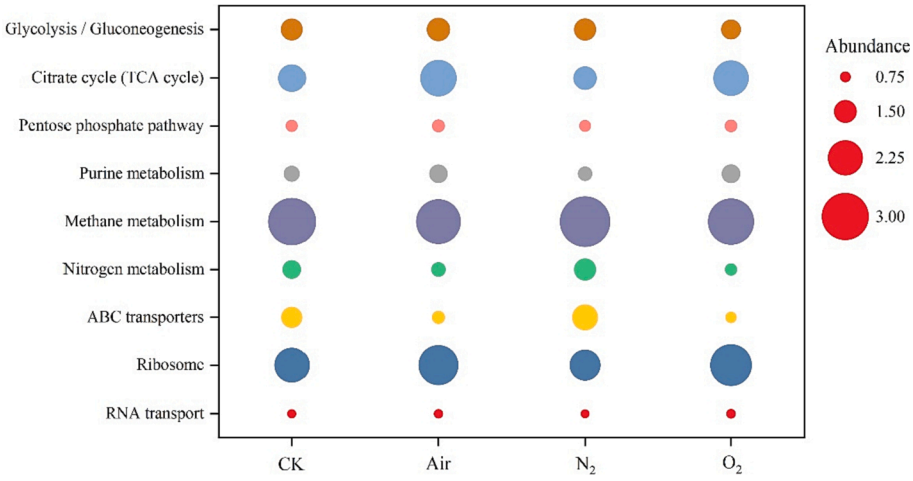


Fig. 6. Abundances of metabolic pathways on Kyoto encyclopedia of genes and genome (KEGG) categories (level 3).

activity of microbial metabolism (Botheju et al., 2009). Reducing oxygen with NADH can release a large amount of energy, which further generates a proton gradient to drive ATP synthesis and transport other metabolites and increase the efficiency of hydrolysis and acidification (Kyzas et al., 2019). In addition, some facultative and aerobic microorganisms can use trace oxygen to completely oxidize acetyl-CoA into CO_2 through aerobic metabolism, such as the TCA cycle (Nguyen and Khanal, 2018). VFAs are not only the products of microbial decomposition of macromolecular organics, but can also be utilized by microbial respiration and metabolism. The VFAs concentration detected in fermentation broth is actually the result of the balance between production and consumption. In general, the abundance of functional genes can reflect the biomass of functional microorganisms in the system. As shown in Fig 6, the high abundance of functional genes can indicate the strong metabolic activity of microorganisms, but it also means that there is a large demand for VFAs as a carbon source. Under O_2-micro-aeration condition, VFAs can be more rapidly consumed for microbial growth and aerobic respiration (Xu et al., 2014), resulting in an increased CO_2 content in biogas considerably (see Supplementary Material).

3. Challenges and perspectives

In recent years, micro-aeration has been investigated as an effective approach to facilitate the performance of AD process and involves controlling VFAs accumulation, stabilizing the process, scavenging H_2S, improving VFAs production, and enhancing hydrolysis and overall methane yield (Nguyen and Khanal, 2018; Chen et al., 2020; Zhang et al., 2021). In this study, the regulation mechanism of micro-aeration for VFAs production was revealed through the comparison of the three gas sources of micro-aeration. In fact, the influence of micro-aeration on the efficiency and performance of VFAs production is related to two aspects: oxygen transfer rate (OTR) and oxygen utilization rate (OUR). The OTR in micro-aeration AD mainly depends on micro-aeration rate, aeration method, reactor configuration, and TS content of substrate (Garcia-Ochoa and Gomez, 2009), and the OUR depends on substrate (type and loading rate) and inoculum used (microbial community) (Díaz et al., 2011; Lim and Wang, 2013; Sawatdeenarunat et al., 2017; Xu et al., 2014). However, OTR and OUR under micro-aerobic conditions with different reactor configurations and operating conditions have yet to be established. In addition, precise automatic online data acquisition and supervisory control systems should be explored in future studies. More practical continuous and long-term experiments with different organic loading rates are needed to further verify the research results.

Research on microbial composition and the behavior of microorganisms are helpful in gaining deeper insights into the activity and biochemical pathways of micro-aeration-based AD for VFAs production and large-scale applications. However, such information is still limited. State-of-the-art techniques, such as meta-omic analyses and stable isotope labeling, should be used in further revealing the microbial community structures, microbial kinetics, genes expression, and metabolic pathways under micro-aeration condition. Future research will be conducted to detect the genes of terminal electron transfer for underlying the mechanism in-depth. For example, *fabG* is associated with the coding of acyl reductases that use NAD^+ or $NADP^+$ as electron acceptor in lipid metabolism and fatty acid synthesis (Zhang et al., 2017); *DesA3* is related to acyl reductase, which acts on paired donors to bind or reduce molecular oxygen directly involved in fatty acid formation (Kachroo et al., 2006).

4. Conclusions

This study revealed the regulation mechanism of micro-aeration for VFAs production from swine manure. VFAs production improved in the first 8 days, and the maximum increased by 22.4% through Air-micro-aeration. Air-micro-aeration at an ORP +5.5 mV higher than the anaerobic ORP was favorable to TVFAs production, while O_2-micro-aeration at an ORP +11.0 mV higher than the anaerobic ORP caused TVFAs reduction due to excess aerobic respiration. The mixing effects of different micro-aeration were limited and no significant change in the bacterial community. Air-micro-aeration enhanced electron transfer activity and microbial metabolism. Archaeal community structure can adjust according to oxygen tolerance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data are available after approval of a research proposal via an established subcommittee.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2022.128140.

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